

Jonathan Daniel Carney

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EDUCATION

- 2021 – present **Ph.D.** in Physics UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL, USA
Advisors: Nicholas Law, Professor of Physics and Astronomy and Igor Andreoni, Assistant Professor of Physics and Astronomy
- 2016 – 2021 **B.S.** in Physics and Mathematics OREGON STATE UNIVERSITY, USA
Thesis title: “Practical Mechanically Interlocked Molecules: A Model of Gas Adsorption in Metal Organic Frameworks Harboring Rotaxane Molecular Shuttles”.
Advisors: Cory Simon, Assistant Professor of Chemical Engineering and David Roundy, Associate Professor of Physics

RESEARCH INTERESTS

- Machine learning for high-cadence time-domain astronomy
- Tidal disruption events
- Optical and high-energy transients
- Computer vision applications in astronomy

HONORS

- 2023 Outstanding Graduate Teaching Assistant Award UNC CHAPEL HILL
2022 Teaching Fellow UNC CHAPEL HILL

PROPOSALS

TELESCOPE TIME (AS PI)

- Gemini-South **2026A** - 5 hours queue time via NOIRLab proposal
- SOAR **2026A** - 18 hours AEON queue time via NOIRLab proposal
- SOAR **2025B** - 36 hours AEON queue time via NOIRLab proposal

OBSERVING EXPERIENCE

- Southern Astrophysical Research (SOAR) Telescope: 10 classical nights (all remote), 2 Gravitational Wave Targets of Opportunity, 1 Einstein Probe Target of Opportunity, weekly AEON queue observations
- Gemini Observatory: 3 queue programs
- Argus Pathfinder Commissioning: 14 nights at the Pisgah Astronomical Research Institute (PARI), 3 nights at the Horace Williams (UNC) Airport

TEACHING EXPERIENCE

INSTRUCTOR OF RECORD

- General Physics II: For Students of the Life Sciences

GRADUATE TEACHING ASSISTANT

- General Physics I: For Students of the Life Sciences
- General Physics II: For Students of the Life Sciences

UNDERGRADUATE TEACHING ASSISTANT

- Paradigms in Physics: Energy and Entropy
- Paradigms in Physics: Quantum Fundamentals
- Paradigms in Physics: Static Fields
- Paradigms in Physics: Oscillations and Waves
- Paradigms in Physics: Central Forces
- Paradigms in Physics: Periodic Systems

CONFERENCES AND WORKSHOPS

CONTRIBUTED TALKS

3. “High-Cadence Transient Discovery with the Argus Array” **Carney, Jonathan;** Corbett, Hank; Law, Nicholas; et al *TDAbench*
2. “Chasing Infant Tidal Disruption Events with SOAR” **Carney, Jonathan;** Andreoni, Igor *Third International Symposium of Science with the SOAR Telescope*
1. “PSF Modeling with Deep Learning for the Argus Optical Array” **Carney, Jonathan;** Corbett, Hank; Marshall, William; Law, Nicholas; Fitton, Shannon; Galliher, Nathan; Gonzalez, Ramses; Machia, Lawrence; Procter, Thomas; Vasquez Soto, Alan *Proc. SPIE 13101, Software and Cyberinfrastructure for Astronomy VIII, 131010O*

POSTER PRESENTATIONS

4. “Optical/Infrared Follow-up of the Longest GRB” **Carney, Jonathan;** Andreoni, Igor; O’Connor, Brendan; et al. *22nd Meeting of the High Energy Astrophysics Division*
3. “PSF Modeling with Deep Learning for Next-Generation Survey Pipelines” **Carney, Jonathan;** Corbett, Hank; Marshall, William; Law, Nicholas; Fitton, Shannon; Galliher, Nathan; Gonzalez, Ramses; Machia, Lawrence; Procter, Thomas; Vasquez Soto, Alan *Rare Gems in Big Data*
2. “PSF Modeling with Deep Learning for Next-Generation Survey Pipelines” **Carney, Jonathan;** Corbett, Hank; Marshall, William; Law, Nicholas; Fitton, Shannon; Galliher, Nathan; Gonzalez, Ramses; Machia, Lawrence; Procter, Thomas; Vasquez Soto, Alan *Hotwiring the Transient Universe VII*
1. “Injecting simulated transient sources for next-generation time-domain survey pipelines with machine learning” **Carney, Jonathan;** Corbett, Hank; Law, Nicholas; Marshall, William; Fitton, Shannon; Galliher, Nathan; Gonzalez, Ramses; Machia, Lawrence; Procter, Thomas; Vasquez Soto, Alan *American Astronomical Society, AAS Meeting 243, id. 261.03*

COMMUNITY OUTREACH

Astronomy on Tap in the Triangle talk: “Artificial Intelligence & the Search for Rare Events”

2025

PUBLICATIONS

REFEREED JOURNAL PUBLICATIONS

(Including arXiv pre-prints)

16. Rose, Kovi; Pritchard, Joshua; Murphy, Tara; et al. (including **Carney, J.**) "Periodic radio and X-ray emission from an accreting white dwarf binary" *Nature Astronomy* (in press), doi:10.1038/s41550-026-02882-x
15. Perley, Daniel A.; Ho, Anna Y. Q.; McGrath, Zoë; et al. (including **Carney, Jonathan**) "AT 2024wpp: an extremely luminous fast ultraviolet transient powered by accretion onto a black hole" *Monthly Notices of the Royal Astronomical Society*, Volume 549, Issue 1, id.stag678
14. Freeburn, James; Dobie, Dougal; Anumalapudi, Akash; et al. (including **Carney, Jonathan**) "Prospects for GRB Afterglow Discovery with the Eric and Wendy Schmidt Observatory System" *arXiv e-prints*, id.arXiv:2604.13650
13. O'Connor, Brendan; Hall, Xander J.; Cabrera, Tomas; et al. (including **Carney, Jonathan**) "SN 2025adpq: A Type Ia supernova in a collisional ring formed during a major galaxy merger" *arXiv e-prints*, id.arXiv:2603.15899
12. Van Bemmell, Natasha; Zhang, Jieli; Cooke, Jeff; et al. (including **Carney, Jonathan**) "An extremely fast fading population II dwarf nova candidate: caught spectroscopically on the rise" *Monthly Notices of the Royal Astronomical Society*, Volume 547, Issue 1, id.stag309
11. Stein, Robert; **Carney, Jonathan**; Ward, Charlotte; et al. "TDE 2025abcr: A Tidal Disruption Event in the Outskirts of a Massive Galaxy" *arXiv e-prints*, id.arXiv:2602.10180
10. Eappachen, D.; Balasubramanian, A.; Swain, V.; et al. (including **Carney, J.**) "Characterizing EP241107a: multiwavelength observations of an Einstein Probe-detected fast X-ray transient" *Monthly Notices of the Royal Astronomical Society*, Volume 545, Issue 1, id.staf2062
9. Swain, Vishwajeet; Ahumada, Tomás; Patil, Sameer K.; et al. (including **Carney, Jonathan**) "GRB 250704B: An Off-axis Short GRB with a Long-lived Afterglow Plateau" *The Astrophysical Journal Letters*, Volume 996, Issue 2, id.L38
8. **Carney, Jonathan**; Andreoni, Igor; O'Connor, Brendan; et al. "Optical/Infrared Observations of the Extraordinary GRB 250702B: A Highly Obscured Afterglow in a Massive Galaxy Consistent with Multiple Possible Progenitors" *The Astrophysical Journal Letters*, Volume 994, Issue 2, id.L46
7. Freeburn, James; Andreoni, Igor; de Soto, Kaylee M.; et al. (including **Carney, Jonathan**) "Identification and Photometric Classification of Extragalactic Transients in the Vera C. Rubin Observatory's Data Preview 1" *The Astrophysical Journal Letters*, Volume 994, Issue 1, id.L24
6. O'Connor, Brendan; Gill, Ramandeep; DeLaunay, James; et al. (including **Carney, Jonathan**) "Comprehensive X-Ray Observations of the Exceptional Ultralong X-Ray and Gamma-Ray Transient GRB 250702B with Swift, NuSTAR, and Chandra: Insights from the X-Ray Afterglow Properties" *The Astrophysical Journal Letters*, Volume 994, Issue 1, id.L17
5. Hu, Lei; Cabrera, Tomás; Palmese, Antonella; et al. (including **Carney, J.**) "Kilonova Constraints for the LIGO/Virgo/KAGRA Neutron Star Merger Candidate S250206dm: GW-MMADS Observations" *The Astrophysical Journal Letters*, Volume 990, Issue 2, id.L46
4. Srinivasaragavan, Gokul P.; Hamidani, Hamid; Schroeder, Genevieve; et al. (including **Carney, J.**) "EP250108a/SN 2025kg: A Jet-driven Stellar Explosion Interacting with Circumstellar Material" *The Astrophysical Journal Letters*, Volume 988, Issue 2, id.L60
3. Galliher, Nathan W.; Procter, Thomas; Law, Nicholas M.; Corbett, Hank; Howard, Ward S.; Vasquez Soto, Alan; Gonzalez, Ramses; Machia, Lawrence; **Carney, Jonathan**; Marshall, William J. "The ArgusSpec Prototype: Autonomous Spectroscopic Follow-up of Flares Detected by Large Array Telescopes"

Publications of the Astronomical Society of the Pacific, Volume 136, Issue 3, id.035004, 18 pp.

2. Corbett, Hank; **Carney, Jonathan**; Gonzalez, Ramses; Fors, Octavi; Galliher, Nathan; Glazier, Amy; Howard, Ward S.; Law, Nicholas M.; Quimby, Robert; Ratzloff, Jeffrey K.; Soto, Alan Vasquez "The Evryscope Fast Transient Engine: Real-time Detection for Rapidly Evolving Transients" *The Astrophysical Journal Supplement Series, Volume 265, Issue 2, id.63, 23 pp.*
1. **Carney, Jonathan**; Roundy, David; Simon, Cory "Statistical thermodynamic model of gas adsorption in a metal-organic framework harboring a rotaxane molecular shuttle" *Langmuir* 2020 36 (43), 13112-13123

CONFERENCE PUBLICATIONS

1. Machia, Lawrence M.; Corbett, Hank; Vasquez Soto, Alan; et al. (including **Carney, Jonathan**) "No photon left behind: image quality optimization for the Argus Optical Array" *Proceedings of the SPIE, Volume 13094, id.1309458, 10 pp.* (2024)

CIRCULARS, TELEGRAMS, & RESEARCH NOTES

39. Hammerstein, E.; Chornock, R.; Sfaradi, I.; **Carney, J.**; Stein, R. "ZTF classification of ZTF26aavrgaw / AT2026nuq and ZTF26aasemes / AT2026kce as likely TDEs" *Transient Name Server AstroNote* 2026-165
38. Hammerstein, E.; Chornock, R.; **Carney, J.**; Sfaradi, I.; Stein, R. "ZTF classification of ZTF26aaqodeo / AT2026irt as a tidal disruption event and reclassification of ZTF25acgtfmh / AT2025ahbd as an AGN" *Transient Name Server AstroNote* 2026-111
37. Hammerstein, E.; Chornock, R.; **Carney, J.**; Stein, R.; Somalwar, J. "ZTF classification of two new TDEs and one Bowen Fluorescence Flare, and spectra/redshifts for three additional TDEs" *Transient Name Server AstroNote* 2026-37
36. Hinds, A. K.; **Carney, J.**; Andreoni, I.; et al. "Tentative classification of AT2026cha as a Rapidly Evolving Type Ic-BL Supernova" *Transient Name Server AstroNote* 2026-31
35. Salgundi, Anirudh; Andreoni, Igor; Anumalapudi, Akash; et al. (including **Carney, Jonathan**) "AT 2026epw: Discovery of a possible nova from Vera C. Rubin Observatory" *The Astronomer's Telegram, No. 17703*
34. Chornock, R.; Hammerstein, E.; Stein, R.; **Carney, J.** "ZTF Classification of AT 2025aljd/ZTF25aclckzl as a likely TDE" *Transient Name Server AstroNote* 2026-7
33. Stein, R.; Yao, Y.; Hall, X. J.; et al. (including **Carney, J.**) "ZTF Identification of ZTF25absjioz (AT 2018mqa) as a possible low-luminosity TDE at 63 Mpc" *Transient Name Server AstroNote* 2025-325
32. Hall, X. J.; Stein, R.; Margutti, R.; et al. (including **Carney, J.**) "X-ray detection of the off-nuclear TDE 2025abcr using Swift" *Transient Name Server AstroNote* 2025-318
31. Stein, R.; **Carney, J.**; Andreoni, I.; et al. "ZTF Classification of ZTF25abxvhmk/AT 2025abcr as a likely offnuclear TDE, possibly from an IMBH, at $z=0.05$ " *Transient Name Server AstroNote* 2025-317
30. Somalwar, J.; Chornock, R.; Margutti, R.; et al. (including **Carney, J.**) "X-ray detection of the TDE 2025aarm using Chandra" *Transient Name Server AstroNote* 2025-316
29. Chornock, R.; Stein, R.; Hammerstein, E.; **Carney, J.** "ZTF Classification of AT 2025aapf/ZTF25abxcnsf as a likely TDE-featureless" *Transient Name Server AstroNote* 2025-309

28. **Carney, Jonathan**; Hall, Xander J.; Andreoni, Igor; O'Connor, Brendan; Busmann, Malte; Gruen, Daniel; Palmese, Antonella "GRB 251026B: SOAR Classification of GOTO25jgl/SN2025absf as a Type Ia supernova" *GRB Coordinates Network Circular* 42519
27. Sears, H.; Blanchard, P. K.; Chornock, R.; et al. (including **Carney, J.**) "GRB 250702B/EP250702a: JWST/NIRCam Observations" *GRB Coordinates Network Circular* 42145
26. Chornock, R.; Hammerstein, E.; Yao, Y.; Margutti, R.; **Carney, J.**; Stein, R.; Graham, M. J. "ZTF Classification of AT 2022aeyz/ZTF22aaqccka as an AGN Turn-on Event in SDSS J002440.72-003019.9" *Transient Name Server AstroNote* 2025-286
25. Hosseinzadeh, G.; Andrews, J.; Sand, D. J.; et al. (including **Carney, J.**) "PASSTA: The Public AEON Spectroscopic Survey for Transient Astronomy" *Transient Name Server AstroNote* 2025-280
24. Stein, R.; Andreoni, I.; **Carney, J.**; Hammerstein, E.; Chornock, R.; Hall, X. J. "ZTF Classification of ZTF25abcemrc/AT2025spf as a likely TDE" *Transient Name Server AstroNote* 2025-251
23. O'Connor, B.; Pasham, D.; Andreoni, I.; et al. (including **Carney, J.**) "GRB 250702B: Chandra X-ray Detection" *GRB Coordinates Network Circular* 41309
22. Dhawan, S.; Goobar, A.; Johansson, J.; et al. (including **Carney, J.**) "Possible bright peculiar/lensed SNIa at $z=0.185$, AT2025ndr" *Transient Name Server AstroNote* 2025-226
21. Sheng, X.; Angus, C. R.; Nicholl, M.; et al. (including **Carney, J.**) "Classification of 2025qhf as a Tidal Disruption Event" *Transient Name Server AstroNote* 2025-223
20. Stein, R.; Das, K.; Chornock, R.; et al. (including **Carney, J.**) "ZTF Classifications of ZTF25aaxdtm (AT2025own) and ZTF25aaoewz (AT 2025iuh) as likely TDEs" *Transient Name Server AstroNote* 2025-212
19. Andreoni, I.; Freeburn, J.; **Carney, J.**; Anumalapudi, A.; Teague, S. "A first collection of extragalactic transient candidates identified in Vera C. Rubin Observatory DP1 data" *Transient Name Server AstroNote* 2025-204
18. Stein, R.; Hammerstein, E.; Chornock, R.; et al. (including **Carney, J.**) "ZTF Classification of ZTF25aaptkov (AT 2025kkj) and ZTF25aaplxiz (AT 2025lxx) as likely TDEs" *Transient Name Server AstroNote* 2025-160
17. Hall, X. J.; Hammerstein, E.; **Carney, J.**; et al. "ZTF/2DTS report of AT2025hbw/ZTF25aajjeon as a featureless TDE candidate at $z=0.628$ " *Transient Name Server AstroNote* 2025-131
16. Hammerstein, E.; Chornock, R.; Stein, R.; et al. (including **Carney, J.**) "ZTF Classification of ZTF25aagfhmz (AT2025cyj), ZTF25aaknwwy (AT2025hee), and ZTF25aagrecw (AT2025gmz) as TDEs" *Transient Name Server AstroNote* 2025-127
15. **Carney, J.**; Stein, R.; Andreoni, I.; et al. "ZTF Classification of ZTF25aajmtzs as a likely TDE at $z=0.136$ " *Transient Name Server AstroNote* 2025-118
14. Hammerstein, E.; Stein, R.; Chornock, R.; Yao, Y.; Andreoni, I.; **Carney, J.** "ZTF report of ZTF25aafwfzz/AT2025cyh as a possible TDE at $z=0.105$ " *Transient Name Server AstroNote* 2025-89
13. Hammerstein, E.; **Carney, J.**; Andreoni, I.; Stein, R.; Chornock, R.; Yao, Y. "ZTF Classification of ZTF25aacnkte/AT2025axp as a likely TDE" *Transient Name Server AstroNote* 2025-74
12. **Carney, J.**; Jacobson-Galan, W.; Andreoni, I.; Schulze, S.; Sollerman, J. "SOAR spectroscopic classification of AT 2025cfc as a Type II supernova with flash features" *Transient Name Server AstroNote* 2025-68

11. Stein, R.; **Carney, J.**; Andreoni, I.; Chornock, R.; Hammerstein, E.; Margutti, R.; Yao, Y. "ZTF Classification of ZTF25aafocs (AT2025bri) as a bright pre-peak TDE" *Transient Name Server AstroNote* 2025-67
10. Das, K. K.; Kasliwal, M. M.; Laz, T. J. D.; et al. (including **Carney, J.**) "ZTF Complete Astronomical Transient Survey within 150 Mpc (CATS150)" *Transient Name Server AstroNote* 2025-64
9. **Carney, J.**; Stein, R.; Hammerstein, E.; Andreoni, I.; Chornock, R.; Yao, Y. "ZTF Classification of ZTF24abwsfdk (AT 2024admt) and ZTF25aabeve (AT 2024agqx) as TDEs" *Transient Name Server AstroNote* 2025-50
8. Freeburn, James; Teague, Sarah; **Carney, Jonathan**; et al. "LIGO/Virgo/KAGRA S250206dm: continued in SOAR galaxy targeted observations and identification of one possible transient" *GRB Coordinates Network Circular* 39286
7. Freeburn, James; **Carney, Jonathan**; Andreoni, Igor; Cook, David "LIGO/Virgo/KAGRA S250206dm: SOAR galaxy targeted observations" *GRB Coordinates Network Circular* 39196
6. Stein, R.; **Carney, J.**; Andreoni, I.; et al. "ZTF Classification of ZTF24abmaujk (AT 2024ymu) as a likely TDE" *Transient Name Server AstroNote* 2024-369
5. Freeburn, J.; Andreoni, I.; **Carney, J.** "GRB 241217A/EP241217b: SOAR afterglow detection" *GRB Coordinates Network Circular* 38642
4. Freeburn, J.; Andreoni, I.; **Carney, J.** "EP241126a: SOAR observations of the optical counterpart" *GRB Coordinates Network Circular* 38409
3. Stein, R.; **Carney, J.**; Andreoni, I.; Earley, N.; Qing, Y.; Hammerstein, E.; Chornock, R.; Ravi, V.; Yao, Y. "Classification of optical transient ZTF24abfaake/AT 2024ule as a tidal disruption event (TDE) at redshift $z=0.112$ " *Transient Name Server AstroNote* 2024-318
2. Freeburn, J.; Andreoni, I.; **Carney, J.** "EP241107a: SOAR upper limits" *GRB Coordinates Network Circular* 38208
1. Freeburn, J.; Andreoni, I.; **Carney, J.** "EP241021a: continued SOAR observations confirm optical rebrightening" *GRB Coordinates Network Circular* 37942

OTHER WORKS

1. Koss, Michael; Aftab, Nafisa; Allen, Steven W.; et al. (including **Carney, Jonathan**) "The Advanced X-ray Imaging Satellite Community Science Book" [textitarXiv:2511.00253](https://arxiv.org/abs/2511.00253)